
Capacity without Access: Reinterpreting the Mid-Depth Spectral Plateau in LLMs

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Abstract

Prior probing-based analyses show that individual layers specialize in distinct linguistic and semantic functions, while a complementary line of work observes that deeper layers enter an alignment-dominated regime where residual transformations become increasingly constrained relative to the evolving hidden-state geometry. These findings appear to conflict: layer-wise functional specialization versus apparent representational stagnation in deeper layers. This raises a key question: Is the observed representational stagnation in deeper layers attributable to suboptimal acquisition or encoding of novel features, or are these features learned appropriately but exhibit minimal marginal contribution to the model’s predictive output? We address this by separating model behavior into (i) representational capacity: the richness and spectral diversity of the encoded features, and (ii) accessibility: the extent to which these features are aligned with, and exploited by, the output-relevant subspace. Analyzing hidden-state second-moment geometry across depth shows that intermediate layers maintain a broad representational span and rich spectral diversity, yet project only weakly onto the output-relevant subspace, indicating intact representational capacity but constrained accessibility within task-relevant dimensions. To probe their functional relevance, we add a controlled diagnostic pathway that routes intermediate-layer embeddings to the terminal readout. Analysis of the induced behavioral changes indicates that the apparent stagnation is better accounted for by geometric limitations on accessibility, rather than by a deficit in representational capacity. The source code is available [here](#).

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1. Introduction

Transformer-based large language models (LLMs) are commonly characterized as constructing hierarchical internal representations, whereby shallower layers predominantly encode surface-level linguistic properties, and progressively deeper layers capture increasingly abstract semantic information (Fan et al., 2024; Ben-Artzy & Schwartz, 2024; Jawahar et al., 2019; Van Aken et al., 2019). This observation indicates that internal representations exhibit continued, substantive development as network depth increases. Concurrently, residual connections function as the primary mechanism for propagating hidden states across layers and for enhancing the stability of the optimization process during training (He et al., 2016). However, recent analyses of residual dynamics demonstrate that parameter updates in deeper network layers become increasingly aligned with the pre-existing internal representations, indicating that the resultant changes are relatively small in magnitude and proceed in a more gradual, incremental fashion (Li & Papyan, 2023; Arefin et al., 2024). This observation naturally gives rise to the following question: if residual updates effect progressively smaller representational changes at greater depths, by what mechanism do deeper layers nonetheless develop increasingly rich semantic structure, as documented in probing-based analyses? Resolving this gap is essential for elucidating the functional role of intermediate-depth representations in LLMs.

To investigate how residual stagnation relates to changes in information routing, we examine the mean attention entropy (MAE) (Ghader & Monz, 2017) to quantitatively characterize how attention distributions evolve across depth in different generations of Llama models (see Fig. 1). MAE serves as a proxy for attention concentration. High entropy indicates a broadly distributed attention pattern, whereas low entropy indicates more concentrated routing over specific tokens. Across the Llama family, MAE is high in early layers and lower in deeper layers, indicating increasingly concentrated routing. Importantly, this trend should not be read as the disappearance of the intermediate entropy peak; rather, the early high-entropy regime appears shorter in later Llama variants. This pattern admits two interpretations: task-relevant structure may emerge earlier, or intermedi-

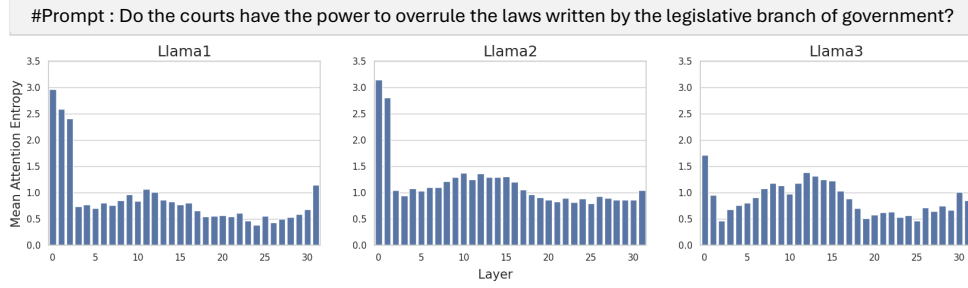


Figure 1. Mean attention entropy (MAE) values across all layers for Llama1,2 and 3, based on a prompt: “Do the courts have the power to overrule the laws written by the legislative branch of government?”. Further examples are provided in Appendix A.

ate layers may still encode rich structure that is accessed through increasingly stabilized routing. Aggregate MAE statistics are provided in Appendix A.1.

However, since MAE is only a coarse proxy for routing, this pattern alone cannot determine whether the earlier concentration of attention reflects efficient processing or representational stagnation, motivating a more direct analysis of the hidden states. To sharpen this analysis, we decompose internal representations along two complementary dimensions: (1) *representational capacity*, defined as the richness of encoded information, operationalized via the diversity of hidden states. (2) *accessibility*, defined as the extent to which this encoded information is utilized by subsequent layers for prediction. This distinction motivates two concrete questions. (Q1) Do successive layers preserve adequate representational capacity as network depth increases? (Q2) If useful features exist in layers, to what extent are they accessible to the terminal prediction pathway?

To address these two questions, we quantify two complementary properties of hidden states. First, we evaluate *representational capacity* using second-moment geometry, employing metrics such as the second-moment spectrum and the effective rank, which collectively characterize the diversity of features encoded within the representation space. Second, we assess *accessibility* by measuring how strongly hidden-state directions align with a readout-relevant subspace induced by the unembedding head.

Subsequently, to test whether mid-depth limitations reflect restricted access rather than insufficient capacity, we introduce the Depth-Aware Diagnostic Intervention (DDI). DDI adds a training-time auxiliary route that aggregates selected intermediate-layer states and delivers them to the terminal readout, strengthening their exposure to the prediction mechanism. The branch is discarded after training, keeping the inference architecture and computation identical to the standard backbone, while enabling us to quantify the contribution of intermediate-layer signals under accessibility. Across models, capacity proxies remain stable in mid-depth (second-moment diversity), while readout cou-

pling increases when an explicit access path is provided, and this accessibility change produces selective behavioral shifts, supporting an under-access interpretation.

2. Related Work

Hierarchical Representations in LLMs. The hierarchical organization of linguistic representations in LLMs has been widely studied using probing methods, which show that lower layers capture surface-level linguistic patterns while deeper layers encode increasingly abstract semantic information (Fan et al., 2024; Ben-Artzy & Schwartz, 2024; Jawahar et al., 2019; Pirozelli et al., 2024; Van Aken et al., 2019; Ju et al., 2024; Jiang et al., 2024; Jin et al., 2025; Sajjad et al., 2022). These findings characterize *what* information is present at different depths, but do not fully explain *how* such information influences the model’s final decisions. Recent techniques such as Tuned Lens (Belrose et al., 2023) and JUMP (Yom Din et al., 2024) show that intermediate layers already contain signals related to the final output, but they do not explicitly address the *geometric constraints* that may prevent these representations from being naturally utilized during inference.

Residual Dynamics and Structural Constraints. A complementary line of research investigates residual dynamics and observes that, in deeper layers, residual updates exhibit increasing alignment with the current hidden state, indicating a progression toward a “near-identity” operating regime (Liu et al., 2020; Li & Pappan, 2023; Arefin et al., 2024; Gerasimov et al., 2025; Barbero et al., 2024; Zhu et al., 2025; Wang et al., 2024). Such behavior has been linked to stable signal propagation across depth. However, the broader implications of this alignment for representation change are not yet fully understood. In particular, prior work primarily examines update vector alignment, overlooking its impact on global geometry and the accessibility of intermediate features to the prediction pathway.

Bridging the Gap. In summary, prior studies have independently examined (i) where different types of linguistic

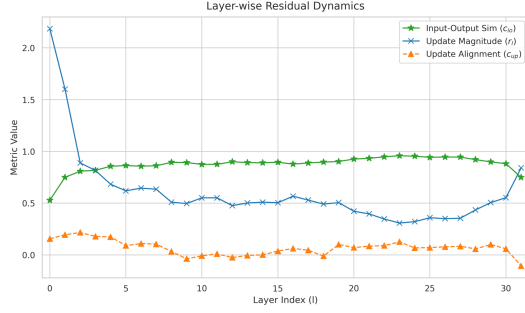


Figure 2. Layer-wise residual update magnitude and input–output similarity exhibit near-identity dynamics in mid-depth layers.

information emerge across depth and (ii) how residual connections support stable signal propagation. However, less is known about how these two aspects interact. Crucially, it remains unclear whether information-rich intermediate representations are reliably propagated to the final output computation. To address this, we jointly analyze representational capacity and accessibility, linking hidden state geometry to functional alignment with output directions.

3. A Local Spectral Analysis

Conceptual Overview. We develop a local spectral view of pre-norm residual dynamics to motivate the distinction between representational capacity and accessibility. In the mid-depth regime, residual updates are empirically small and successive pooled representations remain highly similar (Figure 2). This does not necessarily imply representational collapse: second-moment diversity can remain stable even when coupling to readout-relevant directions is weak. The analysis below formalizes this possibility as a first-order perturbative view; derivations are deferred to Appendix B.

3.1. Pre-norm Residuals as Near-Identity Dynamics

In a pre-norm Transformer, each layer updates its hidden state through a residual addition:

$$H^{(l)} = H^{(l-1)} + F^{(l)}(H^{(l-1)}), \quad (1)$$

where $H^{(l)} \in \mathbb{R}^{T \times d}$ denotes the token-wise hidden states at layer l . To match our empirical protocol in Figure 2, we compute all geometry indicators on pooled sequence representations. Let $\phi : \mathbb{R}^{T \times d} \rightarrow \mathbb{R}^d$ denote the pooling operator used in Figure 2 (e.g., mean-pooling or last-token), and define the token-level residual update tensor

$$U^{(l)} := F^{(l)}(H^{(l-1)}) \in \mathbb{R}^{T \times d}. \quad (2)$$

We then define pooled vectors $h^{(l)} := \phi(H^{(l)})$ and $u_l := \phi(U^{(l)})$. Throughout this subsection, $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ denote

the standard inner product and norm in this pooled vector space (over $h^{(l)}$ and u_l).

Prior analyses suggest that, in deeper layers, the residual function often operates in a near-identity regime (Li & Pappas, 2023; Arefin et al., 2024). We use *near-identity* to mean small *relative* update magnitudes under our pooled protocol, so that $h^{(l)} \approx h^{(l-1)}$ at the pooled level. This is distinct from token-level update–state collinearity.

To capture this behavior at a given input, we decompose the pooled residual update u_l into an input-parallel component and a residual term:

$$u_l = \rho_l h^{(l-1)} + e_l, \quad (3)$$

where ρ_l is the projection coefficient

$$\rho_l := \frac{\langle u_l, h^{(l-1)} \rangle}{\|h^{(l-1)}\|^2}, \quad (4)$$

and e_l denotes the corresponding projection residual (orthogonal to $h^{(l-1)}$ by construction).

To connect this decomposition with observations, we define three layer-wise indicators using the pooled vector u_l :

$$r_l = \frac{\|u_l\|}{\|h^{(l-1)}\|}, \text{ (relative update magnitude)} \quad (5)$$

$$c_{io,l} = \cos(h^{(l-1)}, h^{(l)}), \text{ (input–output similarity)} \quad (6)$$

$$c_{up,l} = \cos(u_l, h^{(l-1)}). \text{ (pooled update alignment)} \quad (7)$$

These indicators are related by definition. Since $c_{up,l} = \frac{\langle u_l, h^{(l-1)} \rangle}{\|u_l\| \|h^{(l-1)}\|}$ and $r_l = \frac{\|u_l\|}{\|h^{(l-1)}\|}$, the coefficient satisfies

$$\rho_l = \frac{\langle u_l, h^{(l-1)} \rangle}{\|h^{(l-1)}\|^2} = c_{up,l} r_l. \quad (8)$$

As shown in Figure 2, distinct phases emerge across depth. In the mid-depth layers, r_l decreases and stabilizes at a small value, while $c_{io,l}$ remains consistently high. Crucially, under our pooled protocol, the update–input alignment $c_{up,l}$ stays near zero rather than approaching one. Because $c_{up,l}$ is computed after pooling, it is not directly comparable to token-level alignment measures used in prior work; it may reflect either consistently small projections or cancellation effects across samples (and thus does not imply token-level orthogonality). Accordingly, the induced parallel coefficient $\rho_l = c_{up,l} r_l$ is small *on average* in this region. Together with the small update magnitude r_l , this pattern is consistent with a regime where residual updates—under our pooled measurement—are too weak to noticeably change the pooled representation across layers. In the final layers, r_l increases while $c_{io,l}$ decreases, which is consistent with the model reshaping representations for the terminal readout.

Table 1. Depth-dependent representational behavior implied by our analysis.

Spectral Observation	Hypothesized functional implication
Stable eigenvalue scale	Capacity preserved (variance directions remain diverse)
Gradual subspace rotation	Potentially weaker coupling to output-relevant directions (access limitation)

3.2. Second-Moment Dynamics

For the second-moment analysis, we interpret the pooled linear map A_l as a distribution-level surrogate (shared across samples) that captures the dominant first-order dependence of $h^{(l)}$ on $h^{(l-1)}$ in the near-identity regime; accordingly, the covariance propagation below should be read as a first-order approximation under this surrogate. We now examine how the second moments of the pooled representation propagate across layers. Under the local linear approximation introduced above, the layer update around the operating point can be written as

$$h^{(l)} \approx A_l h^{(l-1)}, \quad A_l = (1 + \rho_l)I + E_l, \quad (9)$$

where I is the identity and E_l captures local non-parallel deviations. Here, E_l denotes the residual matrix in the pooled, distribution-level linear approximation, rather than the per-sample residual vector e_l introduced earlier. It summarizes the non-parallel first-order components that remain after separating the isotropic term $(1 + \rho_l)I$. The scalar ρ_l is therefore interpreted as a layer-wise coefficient of the isotropic component within this pooled approximation, with the per-sample projection coefficient defined earlier used only as an empirical proxy. The corresponding second-moment matrix $\Sigma_l = \mathbb{E}[h^{(l)}h^{(l)\top}]$ is then approximated by

$$\Sigma_l \approx A_l \Sigma_{l-1} A_l^\top. \quad (10)$$

Expanding this expression and neglecting second-order terms in (ρ_l, E_l) in the small-update regime (e.g., ρ_l^2 and $E_l \Sigma_{l-1} E_l^\top$), we obtain the first-order approximation

$$\Sigma_l \approx \Sigma_{l-1} + 2\rho_l \Sigma_{l-1} + 2\text{sym}(E_l \Sigma_{l-1}). \quad (11)$$

In this decomposition, the term $2\rho_l \Sigma_{l-1}$ modulates the overall scale of Σ_l by uniformly affecting its eigenvalues, whereas the $\text{sym}(E_l \Sigma_{l-1})$ term introduces modest changes to the orientation of the covariance subspace. Hence, ρ_l controls local expansion or shrinkage of the representation, while E_l governs local directional refinement.

In a depth region where residual updates behave as small perturbations (i.e., $|\rho_l| \ll 1$ and $\|E_l\|$ remains moderate), Eq. 11 is consistent with a regime in which the overall second-moment scale changes minimally, while directions retain sufficient flexibility.

3.3. Spectral Interpretation: Capacity, Orientation

We examine the second-moment dynamics from a spectral perspective to understand how residual updates influence

representational scale and orientation. Let $\{\lambda_i^{(l)}\}$ denote the eigenvalues of Σ_l , describing variability along principal representation directions.

(1) Capacity: Stable Spectral Scale. By standard eigenvalue perturbation bounds (e.g., Weyl’s inequality),

$$\left| \lambda_i^{(l)} - (1 + \rho_l) \lambda_i^{(l-1)} \right| \leq \|\Delta \Sigma_l\| \leq C(\|E_l\| + \text{h.o.t.}), \quad (12)$$

for some constant C that depends on $\|\Sigma_{l-1}\|$. Empirically, under layer normalization and our pooling protocol, the scale of Σ_{l-1} remains bounded across depth; moreover, when $|\rho_l| \ll 1$ we have $(1 + \rho_l) \lambda_i^{(l-1)} \approx \lambda_i^{(l-1)}$, yielding the qualitative dependence $|\lambda_i^{(l)} - \lambda_i^{(l-1)}| = O(|\rho_l| + \|E_l\|)$. When residual updates remain small in the mid-depth regime, this bound implies only modest variation in the spectral profile between adjacent layers, which matches our findings that the effective rank and spectrum shape change smoothly rather than collapsing. This suggests that the spectral capacity of the representation is largely preserved within this depth range.

(2) Orientation: Directional Adjustments. Small residual updates do not imply purely input-parallel transformations. In the mid-depth regime, the deviation of E_l primarily induces gradual changes in the orientation of the representational subspace, while its magnitude remains small.

3.4. Implication: Depth-Dependent Constraints

Our local analysis suggests a coherent interpretation of mid-depth dynamics in pre-norm Transformers. Even when residual updates become small and successive hidden states remain highly similar, the representational subspace does not collapse: the second-moment spectrum remains stable and feature diversity is preserved. Instead, the system enters a regime where feature diversity is preserved, but we hypothesize that the pathways through which these features influence downstream processes become constrained.

This decoupling summarizes our structural view: intermediate layers can maintain rich feature representations (capacity), while their contribution to final predictions depends on how effectively these features are routed through depth. This motivates an empirical test: if mid-depth information is preserved yet weakly routed to the terminal readout, can its influence be recovered by modifying only the routing of existing hidden states? To probe this, we introduce *Depth-Aware Diagnostic Intervention (DDI)* in the next section, a controlled training-time intervention that assesses the latent contribution of mid-depth layers without altering the inference-time architecture.

4. Empirical Analysis: Gradient-Guided Access Reconfiguration

We test whether mid-depth features remain capacity-preserving but under-accessed, and whether an explicit training-time route increases their readout-side influence.

4.1. Why External Readout Is Required

As demonstrated in Section 3, the model exhibits a mid-depth, near-identity transformation profile. The update magnitude remains small, the similarity between inputs and outputs is high, and the pooled update–input alignment c_{up} remains approximately zero. This suggests that mid-depth layers may retain representational capacity, while their influence on final predictions is constrained by residual routing.

Separating routing effects from other Transformer components is difficult, since attention, MLPs, and normalization jointly shape the final representation. We therefore introduce an auxiliary depth-aware readout (DDI) that provides a training-time gradient pathway from intermediate layers to the terminal representation. We jointly optimize the backbone and the DDI pathway with an intervention strength γ so the residual stream remains dominant. This enables gradients to flow through an explicit mid-depth route and tests whether improved access yields behavioral changes, while leaving the inference-time architecture unchanged.

4.2. Depth-Aware Diagnostic Intervention (DDI)

To diagnose structural constraints on representational accessibility, we introduce a **Depth-Aware Diagnostic Intervention (DDI)** module. Unlike static probing, DDI constructs a *trainable* depth-aware summary vector $R_t \in \mathbb{R}^d$ from mid-depth activations and adds a small auxiliary readout term at the terminal representation, creating an explicit access route from mid-depth states to the terminal readout.¹

$$\tilde{h}_t^{(L)} = h_t^{(L)} + \gamma R_t, \quad t = 1, \dots, T, \quad \gamma \ll 1, \quad (13)$$

where $h_t^{(L)} \in \mathbb{R}^d$ is the original final-layer hidden state at token t , and $\tilde{h}_t^{(L)}$ is the modified representation. The scalar γ controls the intervention strength, ensuring that DDI acts as a small perturbation while enabling gradients to propagate through the auxiliary route defined by R_t .

Figure 3 summarizes the training-time role of DDI; detailed architectural specifications, aggregation procedures, and training objectives are provided in Appendix F. DDI is designed to isolate readout-side accessibility under three constraints.

Internal-state-only Construction. DDI introduces no external knowledge, retrieval, or memory. The diagnostic

vector R_t is computed from intermediate hidden states already produced by the backbone, so the intervention uses only internally available information.

Training-time Readout Route. With intervention strength γ , DDI provides an auxiliary route that couples selected intermediate directions to the terminal readout. This tests whether strengthening an explicit readout-facing path for mid-depth structure induces measurable representational and behavioral changes.

Removal after Internalization. After the diagnostic stage, we gradually remove the additive term in Eq. 13 while restoring the original architecture. We encourage the backbone final-layer representation $H^{(L)}$ to match the DDI-augmented representation $\tilde{H}^{(L)}$ via self-distillation with auxiliary objectives (Appendix F). Thus, DDI adds no external information or inference-time capacity; it provides a temporary training-time route that can increase the alignment between relatively preserved intermediate representations and the terminal readout.

Measuring Accessibility via the Unembedding-induced Subspace. To operationalize *accessibility* as readout-relevant coupling, we measure how strongly each layer’s representation projects onto a fixed subspace induced by the unembedding head. Let $W_U \in \mathbb{R}^{|V| \times d}$ denote the unembedding matrix of the *pre-intervention base* backbone. We form $W_s \in \mathbb{R}^{m \times d}$ by sampling m rows of W_U and define $B_{out} \in \mathbb{R}^{d \times k}$ as the top- k eigenspace of $W_s^\top W_s$, with projector $P_{out} = B_{out} B_{out}^\top$. Given a pooled layer representation $h^{(l)} = \phi(H^{(l)}) \in \mathbb{R}^d$ (same pooling protocol as Section 3), we define the layer-wise access ratio

$$A_l = \frac{\mathbb{E} [\|P_{out} h^{(l)}\|_2^2]}{\mathbb{E} [\|h^{(l)}\|_2^2]}, \quad P_{out} = B_{out} B_{out}^\top. \quad (14)$$

We compute B_{out} once from the pre-intervention base backbone and hold it fixed when comparing baseline vs. internalized models. Implementation details are provided in Appendix D.

4.3. Second-Moment Responses: Structural Adjustment without Capacity Expansion

When gradient-based rerouting and internalization are enabled via DDI, the model exhibits structural modifications without observable expansion of global representational capacity (Figures 4, 5, 6, 7, 8, 9, 10), as quantified through multiple representation metrics including CKA (Kornblith et al., 2019), Effective Rank (Roy & Vetterli, 2007) & Log-Det (Yu et al., 2020), and SVCCA (Raghu et al., 2017).

CKA Difference Map (Figure 4). The CKA difference map shows *reduced* similarity between the terminal layer

¹Full architectural details of DDI are provided in Appendix F.

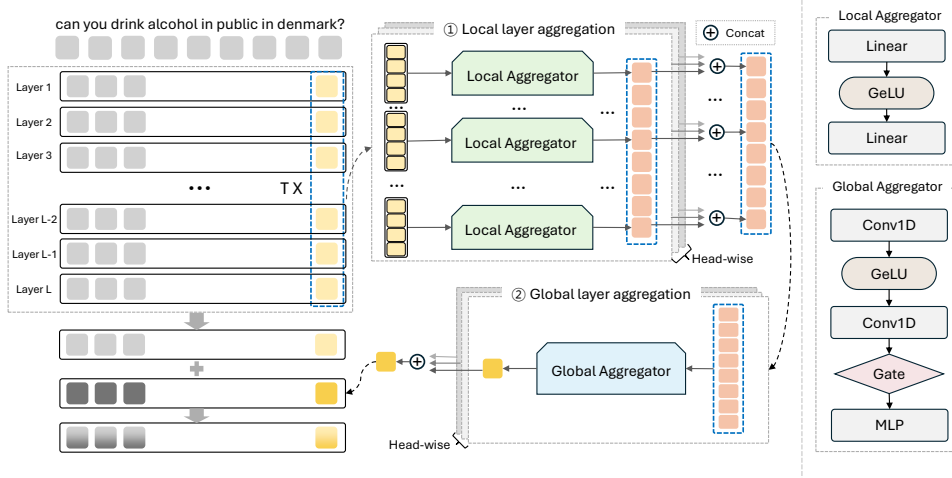


Figure 3. Overview of the Depth-Aware Diagnostic Intervention (DDI). During training, DDI aggregates intermediate hidden states into a depth-aware summary and exposes this summary to the terminal readout through a small additive route. After internalization, the auxiliary route is removed and the finalized backbone is evaluated through the original inference path.

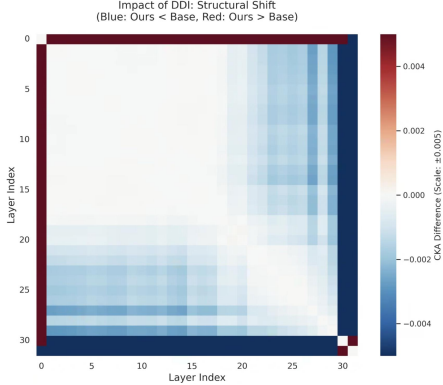


Figure 4. CKA comparison between the baseline backbone and the DDI-trained backbone (after internalization, evaluated with DDI removed), shown as a layer-wise similarity/difference map.

and earlier layers. Since CKA compares the centered sample-sample similarity (Gram) structure across layers, this decrease indicates that DDI changes how the terminal layer globally organizes examples in representation space, i.e., a global reorganization rather than a uniform preservation of the original inter-layer geometry.

Effective Rank and Log-Det (Figures 5, 6). Despite structural shifts, the rank plateau and volume reduction in the mid-depth region remain close to those of the residual-only baseline. This indicates that the intervention does not artificially inflate representational capacity; instead, adjustments occur within the expressive range of the network.

SVCCA Alignment (Figures 7, 8). In contrast to CKA’s global measure, SVCCA captures the most correlated shared

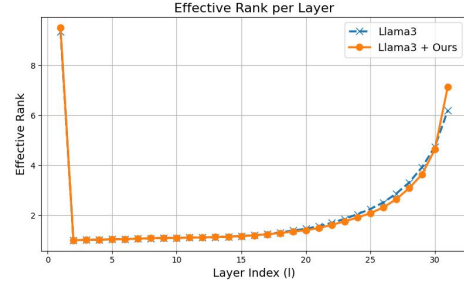


Figure 5. Layer-wise Effective Rank comparison between Llama3 and Llama3+Ours.

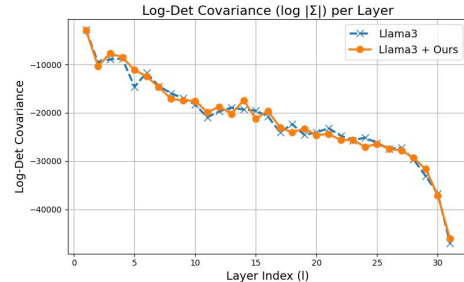


Figure 6. Log-Det second-moment matrix across layers in Llama3 vs. Llama3+Ours.

linear subspace between layers. Under DDI, SVCCA increases between final and shallow/mid-depth layers, suggesting a stronger shared subspace across depth despite overall reorganization indicated by CKA ($\downarrow$). Meanwhile, unembedding-subspace access becomes pronounced toward readout-facing layers after internalization (Fig. 9), supporting the interpretation that DDI strengthens readout coupling.

Mean Attention Entropy (Figure 10). As an auxiliary indicator, MAE moderately increases in the mid-to-deep

Table 2. Commonsense reasoning. “+Ours” denotes the finalized backbone after internalization, evaluated with DDI removed. Averaged over five seeds; Avg. reports mean and standard deviation, and per-dataset standard deviations are provided in Appendix E.1.

Method	OBQA	ARC-e	ARC-c	WinoG.	PIQA	BoolQ	HellaS.	Avg.
Llama3 + LoRA	84.9	90.2	79.9	85.5	87.9	69.9	94.4	84.7±0.2
+ Ours	85.3 (+0.4)	90.3 (+0.1)	80.5 (+0.6)	85.1 (-0.4)	88.7 (+0.9)	71.8 (+1.9)	94.9 (+0.5)	85.2±0.2 (+0.5)
Qwen2 + LoRA	89.3	92.7	83.1	85.5	89.5	73.9	94.6	86.9±0.3
+ Ours	89.1 (-0.2)	93.2 (+0.5)	84.2 (+1.1)	85.7 (+0.2)	89.8 (+0.3)	74.1 (+0.2)	94.8 (+0.2)	87.3±0.2 (+0.4)
Mistral + LoRA	84.0	85.7	73.4	83.0	87.4	71.0	90.3	82.1±0.5
+ Ours	85.4 (+1.4)	86.1 (+0.4)	74.6 (+1.2)	84.5 (+1.5)	87.5 (+0.1)	71.8 (+0.8)	90.7 (+0.4)	82.9±0.6 (+0.8)
Phi-3.5 + LoRA	88.6	94.4	85.4	83.0	86.1	70.3	90.8	85.5±0.3
+ Ours	89.3 (+0.7)	94.7 (+0.3)	86.0 (+0.6)	83.6 (+0.6)	86.8 (+0.7)	70.5 (+0.2)	90.8 (±0.0)	86.0±0.4 (+0.5)

Table 3. Arithmetic reasoning. “+Ours” denotes the finalized backbone after internalization, evaluated with DDI removed. Averaged over five seeds; Avg. reports mean and standard deviation, and per-dataset standard deviations are provided in Appendix E.1.

Method	SVAMP	AddSub	AQuA	MultiArith	GSM8K	Avg.
Llama3 + LoRA	82.7	92.9	32.3	98.0	72.9	75.8±0.6
+ Ours	83.8 (+1.1)	92.9 (±0.0)	34.3 (+2.0)	98.3 (+0.3)	73.6 (+0.7)	76.6±0.4 (+0.8)
Qwen2 + LoRA	82.8	91.6	36.9	97.3	73.9	76.5±0.7
+ Ours	83.5 (+0.7)	91.2 (-0.4)	37.6 (+0.7)	97.3 (±0.0)	75.0 (+1.1)	76.9±0.6 (+0.4)
Mistral + LoRA	65.3	87.9	23.7	95.0	62.6	66.9±1.5
+ Ours	66.8 (+1.5)	88.9 (+1.0)	24.7 (+1.0)	95.3 (+0.3)	63.2 (+0.6)	67.8±1.0 (+0.9)
Phi-3.5 + LoRA	76.4	86.1	33.6	98.1	76.0	74.0±0.3
+ Ours	76.9 (+0.5)	85.5 (-0.6)	34.1 (+0.5)	98.4 (+0.3)	77.0 (+1.0)	74.4±0.3 (+0.4)

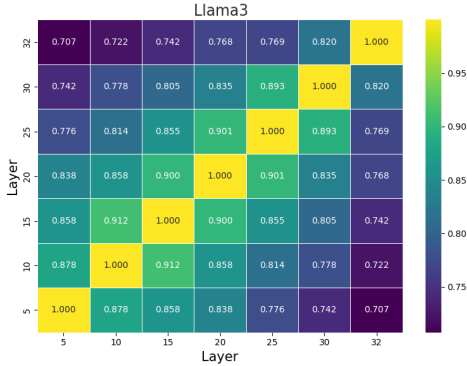


Figure 7. Layer-wise SVCCA in Llama3.

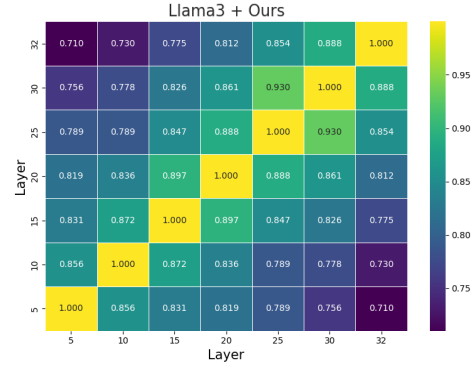


Figure 8. Layer-wise SVCCA in Llama3 + Ours.

region after intervention, suggesting less peaked attention and a modest broadening of attention patterns.

Additional Controls for Interpretive Specificity. The preceding analyses indicate that DDI changes readout-side coupling while leaving broad capacity-oriented metrics nearly unchanged. We next examine whether this pattern can instead be explained by broad representational rewriting or by a metric-specific artifact.

First, we compare both LoRA and Ours against the same pretrained backbone. As shown in Table 4, Ours remains as close as, or closer than, LoRA to the pretrained representation under CKA while also showing higher access ratio at representative layers. This supports accessibility-side realignment rather than large-scale representational rewriting.

Second, we examine whether the MAE increase conflicts with the small change in effective rank. Under our decom-

Table 4. Pretrained-anchor comparison on Llama3. Ours remains close to the pretrained backbone under CKA while increasing readout access ratio (AR), suggesting accessibility-side realignment rather than broad representational rewriting.

Layer	CKA(Pre, LoRA)	CKA(Pre, Ours)	AR(Pre)	AR(LoRA)	AR(Ours)
7	0.919	0.943	0.344	0.359	0.365
17	0.930	0.934	0.339	0.350	0.363
27	0.860	0.893	0.347	0.343	0.355

position, it does not: effective rank and participation ratio measure broad second-moment richness, whereas A_l measures readout-side coupling. Table 5 shows that changes remain small in the plateau region but become clearer near the final readout-facing layers. Participation ratio follows the same qualitative tendency as effective rank, suggesting that the interpretation does not depend on a single second-order geometry metric.

Overall, these results support the interpretation that DDI

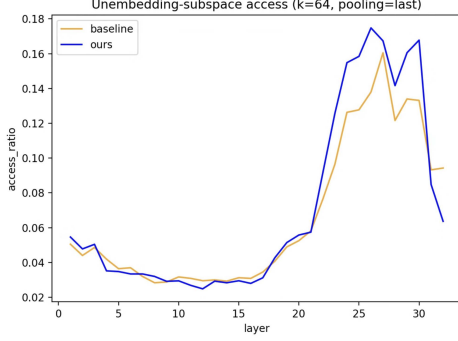


Figure 9. Layer-wise unembedding-subspace access ratio A_l (Eq. 14) for baseline vs. internalized (+Ours) models.

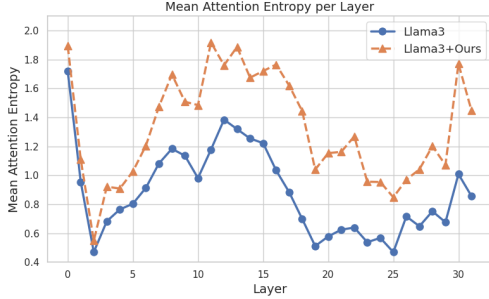


Figure 10. Layer-wise MAE in Llama3 and Llama3+Ours. Additional examples are provided in Appendix A.

does not broadly expand representational capacity, but selectively changes how existing intermediate structure is coupled to the terminal readout. The combination of stable capacity-oriented metrics, increased readout access, and preserved pretrained-anchor similarity is consistent with accessibility-side reorganization. Additional results for other model families are provided in Appendix G.4.

4.4. From Under-Formation to Under-Access

These results support the view that mid-depth layers possess sufficient representational capacity, as evidenced by stable second-moment statistics and rank profiles. In the residual-only baseline, however, this capacity can remain *under-accessed*: representations become overly stabilized, reflected by high inter-layer similarity (CKA) together with small pooled updates, limiting their contribution to the terminal prediction mechanism. Through learning, DDI alleviates this *directional misalignment*, improving the degree to which mid-depth information influences the final layer without changing second-moment capacity metrics. Thus, the mid-depth plateau induced by residual alignment is better characterized not as a failure of representation formation but as a regime where access pathways are partially constrained. This suggests that residual connections serve not only as an optimization-stabilizing mechanism but also as a structural

Table 5. Depth-localized changes in Llama3. Capacity-oriented metrics remain nearly unchanged in the plateau region, while access-ratio changes become clearer near the readout-facing layers.

Metric	Plateau Δ (L10–L12)	Final Δ (L29–L31)	Contrast
Effective rank	+0.0011	+0.1509	+0.1498
Participation ratio	-0.0005	+0.0335	+0.0330
Access ratio (A_l)	-0.00240	+0.01397	+0.01637

factor that can restrict the utilization of representations.

5. Experimental Validation: Behavioral Sensitivity to Access Reconfiguration

This section examines how residual-based structural restrictions, identified in Section 4, manifest in the behavioral output of LLMs. Our focus is not on achieving performance gains, but on testing the *access restriction hypothesis* by observing whether partially restored accessibility leads to selective behavioral differences across task complexity.

5.1. Setup

Models and Training. We evaluate four representative LLMs: Llama-3-8B (Dubey et al., 2024), Qwen2-7B (Hui et al., 2024), Mistral-7B (Jiang et al., 2023), and Phi-3.5-Mini (Abdin et al., 2024). Training follows the two-stage protocol in Section 4.2. All results are reported on the finalized backbone with DDI removed (zero inference overhead). Unless stated, the baseline denotes the LoRA-fine-tuned model, and “+Ours” denotes the LoRA-fine-tuned model trained with DDI and evaluated after internalization with the auxiliary route removed. The untouched pretrained model is reported only as supplementary context when explicitly indicated. Training details, hyperparameters, and datasets are provided in Appendix C. Run-to-run variance is summarized in the Avg. column of the main result tables, and per-dataset standard deviations are reported in Appendix E.1.

Evaluation Tasks. We group benchmarks by reasoning demands. For commonsense and logical pattern matching, we use OpenBookQA (OBQA) (Mihaylov et al., 2018a), ARC-E/ARC-C (Clark et al., 2018b), Winogrande (Sakaguchi et al., 2020), PIQA (Bisk et al., 2020), BoolQ (Clark et al., 2019b), and HellaSwag (Zellers et al., 2019b). For arithmetic and multi-hop reasoning, we use SVAMP (Patel et al., 2021a), AddSub (Hosseini et al., 2014b), AQuA (Ling et al., 2017b), MultiArith (Roy & Roth, 2015b), and GSM8K (Cobbe et al., 2021b). Results are averaged over five runs with different seeds under identical settings.

5.2. Task-Dependent Behavioral Sensitivity

When access reconfiguration is applied, we observe task-dependent behavioral changes across both commonsense

Table 6. Control experiment on structural efficacy.

Method	Structure Type	Param Budget	Avg (%)
Baseline (LoRA)	-	-	75.7
Depth-Unaware Control	Mean Pooling	$\approx 15\text{M}$	75.8
Simple Depth-Aware Control	Weighted Sum	$\approx 15\text{M}$	75.8
Depth-Aware (Ours)	Hierarchical	$\approx 15\text{M}$	76.6

and arithmetic benchmarks (Tables 2, 3), with relatively larger gains often appearing on benchmarks that plausibly involve more multi-step or abstraction-heavy reasoning.

(A) Sensitivity in Complex Reasoning. On arithmetic and multi-step reasoning benchmarks (e.g., GSM8K, SVAMP, and AQuA), we observe consistent but moderate improvements across models (Avg. gains of roughly $+0.4$ – $+0.9$ points, with per-dataset gains up to $+2.0$). We observe a similar sensitivity on more challenging commonsense-style benchmarks that stress non-trivial inference (notably ARC-C). On some other commonsense tasks (e.g., OBQA and WinoGrande), the effect is more mixed across model families despite positive average shifts. Overall, this pattern is compatible with the view that mid-depth features are present but may not be fully reflected in the terminal readout under a residual-only pathway. The positive shifts are consistent with the interpretation that improving accessibility can increase the extent to which latent information is leveraged by the prediction mechanism.

(B) Insensitivity in Pattern-driven Tasks. On more structured or pattern-driven arithmetic benchmarks such as AddSub and MultiArith, improvements are generally smaller. Likewise, on several commonsense benchmarks that may be well supported by the backbone’s primary computation pathway (e.g., ARC-E, PIQA, BoolQ, HellaS), gains are often below ~ 0.5 points, though the effect varies by dataset and model. This suggests that some tasks may already be adequately handled by the residual pathway, leaving less headroom for accessibility-driven shifts, while other tasks can benefit from improved access. Overall, the behavioral effect of access reconfiguration is not uniform across tasks and appears to depend on the evaluation setting.

5.3. Representation–Behavior Alignment

The representation and behavior results are consistent with the access-bottleneck interpretation. Capacity-oriented metrics remain largely stable, whereas DDI reorganizes the terminal representation ($\text{CKA}\downarrow$) and increases readout-side coupling through shared subspaces and output-relevant directions ($\text{SVCCA}\uparrow$, $A_l\uparrow$). The behavioral gains are correspondingly more visible on arithmetic and harder commonsense benchmarks, where prediction is more sensitive to the recruitment of intermediate structure.

5.4. Structural Control: Layer-wise Importance

To verify that the observed gains are not simply due to additional parameters or a generic auxiliary pathway, we run parameter-matched controls throughout all experiments that simplify the depth structure. We compare DDI with a depth-unaware mean-pooling route and a simple learned weighted-sum route over layers. Both controls perform close to the LoRA baseline, indicating that the improvement is not captured by parameter count, an auxiliary route, or simple linear depth aggregation alone.

6. Discussion

Our results support viewing the mid-depth plateau as an access bottleneck rather than a simple failure of representation formation. DDI is intended as a diagnostic route for studying this accessibility effect, not as a general-purpose fine-tuning architecture. Since internalization updates the backbone, the results should be interpreted as evidence for improved readout-side alignment of relatively preserved intermediate structure, rather than as direct proof that all task-relevant features were fully formed before fine-tuning.

Scope and Architectural Dependence. Our study is limited to small-to-mid-scale, PreLN-style LLMs under a LoRA-based fine-tuning regime. Normalization placement, residual routing, attention routing, and optimizer-induced representation allocation may alter the balance between formation and accessibility. Related alternatives such as Mix-LN (Li et al., 2024), Muon-related optimizer analyses (Jha & Reagen, 2026), and lambda-skip connections (Joseph et al., 2024) can therefore be viewed as complementary ways to modify this balance. Direct comparisons to these alternatives, as well as head-level routing analysis and selective pathway manipulation, remain important directions for future work. See Appendix G for ablation studies and Appendix G.3 for qualitative examples.

7. Conclusion

We examine the mid-depth spectral plateau phenomenon and show that it may be interpreted not only as limited representational formation but also as a structural restriction on *accessibility* to the terminal readout. Together with depth-aware intervention, our results support the hypothesis that mid-depth representations retain rotational flexibility but may contribute weakly to terminal prediction under residual-only pathways. These findings should be read as diagnostic evidence for separating representation formation from readout-side utilization, rather than as a claim of uniform practical gains across models or tasks. Beyond asking *what* is encoded, future work should consider *how* formed representations participate in the final decision process.

Acknowledgements

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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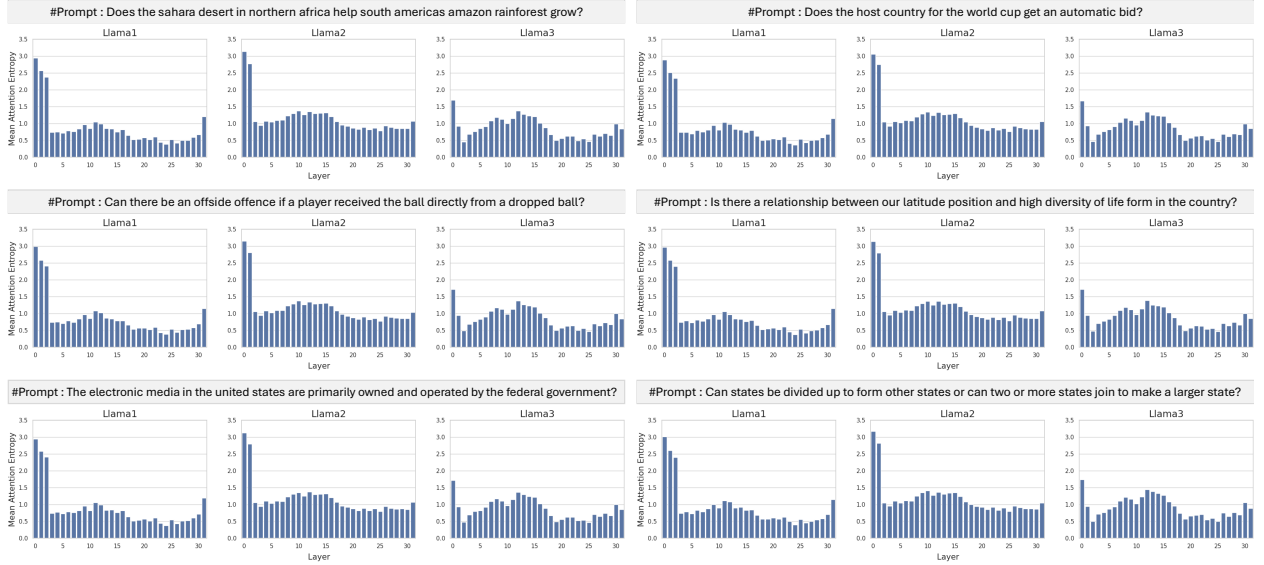


Figure 11. Mean attention entropy (MAE) across all layers for Llama1, Llama2, and Llama3.

A. Mean Attention Entropy Results from Various Prompt Examples

We visualize additional mean attention entropy (MAE) results obtained from various prompt examples. Figure 11 compares the MAE values of Llama1, Llama2, and Llama3, while Figure 12 shows the comparison between Llama3 and our proposed Llama3+Ours model.

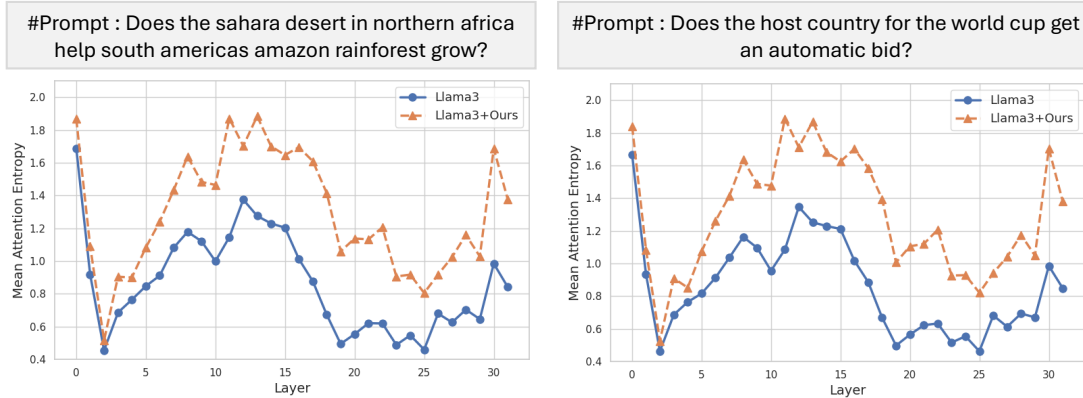


Figure 12. Mean attention entropy (MAE) comparison between Llama3 and Llama3+Ours (example 1).

A.1. Aggregate MAE over ARC-Challenge

To avoid relying on a few qualitative prompts, we compute aggregate MAE over the full ARC-Challenge test set. The secondary peak remains visible after averaging, rather than being a prompt-specific artifact. Thus, our interpretation is not that the intermediate peak disappears, but that MAE decreases after the initial early-layer peak, with the earliest high-entropy regime becoming shorter in later models.

B. Perturbative Analysis of Residual Dynamics

In this section, we provide a theoretical framework to analyze the structural constraints identified in Section 3. Our analysis relies on **first-order perturbation theory**, treating the residual updates as local linear surrogates. Note that the derivations below are intended as a local characterization of the mid-depth regime where updates are empirically small, rather than a

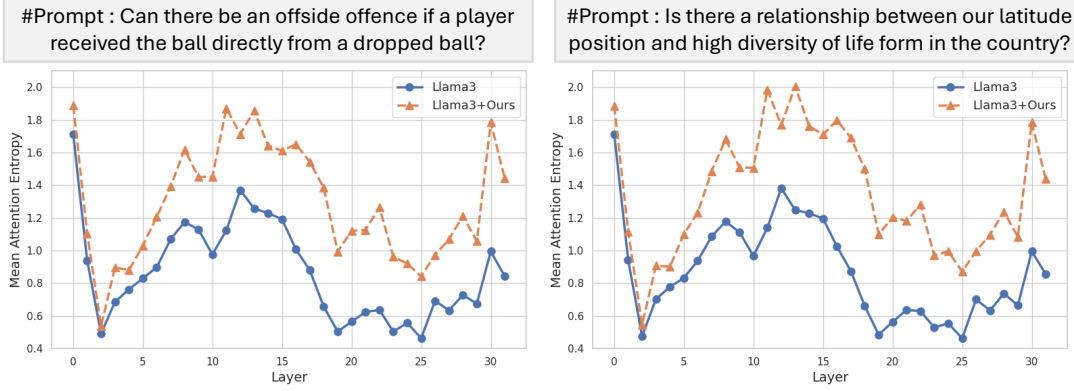


Figure 13. Mean attention entropy (MAE) comparison between Llama3 and Llama3+Ours (example 2).

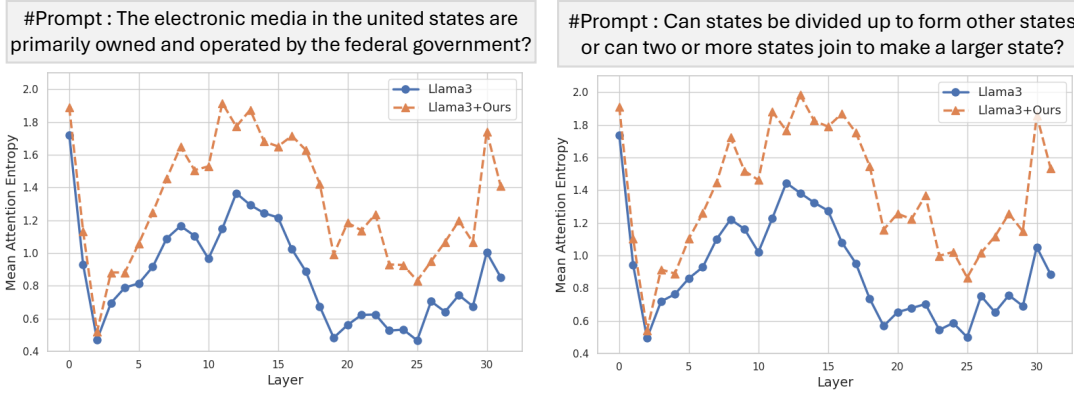


Figure 14. Mean attention entropy (MAE) comparison between Llama3 and Llama3+Ours (example 3).

global description of the entire Transformer dynamics.

B.1. Setup and Assumptions

We follow the pooled measurement protocol of Section 3. Let $H^{(l)} \in \mathbb{R}^{T \times d}$ be the token-wise hidden states and define pooled vectors $h^{(l)} := \phi(H^{(l)}) \in \mathbb{R}^d$ for the pooling operator ϕ used in Fig. 2. Let $U^{(l)} := F^{(l)}(H^{(l-1)}) \in \mathbb{R}^{T \times d}$ denote the token-wise residual update, and define the pooled residual increment proxy $u_l := \phi(U^{(l)})$. When ϕ is linear (e.g., mean-pooling or last-token), $h^{(l)} = h^{(l-1)} + u_l$ holds exactly; otherwise we use it as a pooled increment proxy.

Assumption 1 (Pooled Local Linear Surrogate). In the near-identity regime under our pooled protocol (layers where $r_l \ll 1$), we use a first-order pooled surrogate in which the pooled increment can be approximated by a linear operator acting on the pooled input:

$$u_l \approx B_l h^{(l-1)}, \quad (15)$$

where B_l denotes an *effective* linear operator capturing the dominant first-order dependence of the pooled increment on the pooled representation over the empirical distribution. Equivalently, the pooled representation follows

$$h^{(l)} \approx (I + B_l) h^{(l-1)}. \quad (16)$$

This surrogate is not assumed to hold exactly; it is introduced solely to derive first-order implications in the empirically small-update regime.

Assumption 2 (Second-Moment Convention). We analyze the uncentered second moment $\Sigma_l := \mathbb{E}[h^{(l)} h^{(l)\top}]$ to match the pooled geometry protocol used in the main text.

Table 7. Aggregate MAE across 1,172 ARC-Challenge prompts.

Model	# Prompts	L1	L2	L10	L11	L12	L30	L31
Llama3	1172	2.10	0.95	1.81	1.46	1.85	0.80	1.37
Llama3 + Ours	1172	2.12	1.16	1.99	1.69	2.10	1.18	1.73

B.2. Derivation of Second-Moment Dynamics

Notation. Throughout this appendix, $\|\cdot\|_2$ denotes the spectral norm. We use the uncentered second moment $\Sigma_l := \mathbb{E}[h^{(l)}h^{(l)\top}]$, matching the geometry indicators in the main text.

Proposition B.1 (First-Order Second-Moment Update (Pooled Surrogate)). *Assume we approximate the pooled update by a local linear surrogate of the form $h^{(l)} \approx A_l h^{(l-1)}$ under the pooled protocol. Further decompose the pooled update surrogate as in the main text,*

$$A_l = (1 + \rho_l)I + E_l,$$

where ρ_l is the pooled input-parallel coefficient defined in Section 3 and E_l captures the remaining non-parallel deviation at the pooled level. Then the uncentered second moment admits the expansion

$$\Sigma_l \approx A_l \Sigma_{l-1} A_l^\top = \Sigma_{l-1} + 2\rho_l \Sigma_{l-1} + 2 \text{sym}(E_l \Sigma_{l-1}) + O((\rho_l^2 + |\rho_l| \|E_l\|_2 + \|E_l\|_2^2) \|\Sigma_{l-1}\|_2),$$

where $\text{sym}(A) = \frac{1}{2}(A + A^\top)$.

Proof. By the surrogate assumption,

$$\Sigma_l = \mathbb{E}[h^{(l)}h^{(l)\top}] \approx \mathbb{E}[A_l h^{(l-1)}h^{(l-1)\top} A_l^\top] = A_l \Sigma_{l-1} A_l^\top.$$

Substituting $A_l = (1 + \rho_l)I + E_l$ and expanding yields

$$\Sigma_l = (1 + \rho_l)^2 \Sigma_{l-1} + (1 + \rho_l)(E_l \Sigma_{l-1} + \Sigma_{l-1} E_l^\top) + E_l \Sigma_{l-1} E_l^\top.$$

We use $(1 + \rho_l)^2 = 1 + 2\rho_l + O(\rho_l^2)$ and $(1 + \rho_l) = 1 + O(\rho_l)$ and collect all second-order contributions (including ρ_l^2 , $\rho_l(E_l \Sigma_{l-1} + \Sigma_{l-1} E_l^\top)$, and $E_l \Sigma_{l-1} E_l^\top$) into the remainder term. The dependence on $\|\Sigma_{l-1}\|_2$ follows from submultiplicativity of the spectral norm, e.g.,

$$\|E_l \Sigma_{l-1}\|_2 \leq \|E_l\|_2 \|\Sigma_{l-1}\|_2, \quad \|E_l \Sigma_{l-1} E_l^\top\|_2 \leq \|E_l\|_2^2 \|\Sigma_{l-1}\|_2.$$

This gives

$$\Sigma_l = \Sigma_{l-1} + 2\rho_l \Sigma_{l-1} + (E_l \Sigma_{l-1} + \Sigma_{l-1} E_l^\top) + O((\rho_l^2 + |\rho_l| \|E_l\|_2 + \|E_l\|_2^2) \|\Sigma_{l-1}\|_2).$$

Noting $E_l \Sigma_{l-1} + \Sigma_{l-1} E_l^\top = 2 \text{sym}(E_l \Sigma_{l-1})$ completes the proof. Dropping the remainder yields the same first-order form as Eq. 11 in the main text. $\square$

Remark (Interpretation at the Pooled Level). The decomposition $A_l = (1 + \rho_l)I + E_l$ is a pooled-level local surrogate, intended to summarize near-identity behavior under the measurement protocol of Fig. 2. In particular, ρ_l corresponds to the pooled projection coefficient derived from $u_l = \rho_l h^{(l-1)} + e_l$ and satisfies $\rho_l = c_{up,l} r_l$. Because ρ_l is computed after pooling, it should not be interpreted as a token-level collinearity parameter.

B.3. Analysis of Spectral Stability

Proposition B.2 (Eigenvalue Perturbation Bound). *Let $A := (1 + \rho_l)^2 \Sigma_{l-1}$ and $P := 2 \text{sym}(E_l \Sigma_{l-1})$ under the pooled surrogate. Then the eigenvalues of Σ_l deviate from the uniformly scaled spectrum by*

$$|\lambda_i^{(l)} - (1 + \rho_l)^2 \lambda_i^{(l-1)}| \leq 2 \|\Sigma_{l-1}\|_2 \|E_l\|_2 + O((\rho_l^2 + |\rho_l| \|E_l\|_2 + \|E_l\|_2^2) \|\Sigma_{l-1}\|_2).$$

Proof. From Proposition 1, we write Σ_l as a perturbation of the scaled prior second moment:

$$\Sigma_l = \underbrace{(1 + \rho_l)^2 \Sigma_{l-1}}_A + \underbrace{2 \text{sym}(E_l \Sigma_{l-1})}_P + O((\rho_l^2 + |\rho_l| \|E_l\|_2 + \|E_l\|_2^2) \|\Sigma_{l-1}\|_2).$$

Here A is symmetric since Σ_{l-1} is symmetric, and P is symmetric by definition of $\text{sym}(\cdot)$. Thus we can invoke Weyl’s inequality: for symmetric A and P ,

$$|\lambda_i(A + P) - \lambda_i(A)| \leq \|P\|_2.$$

The spectral norm of P is bounded by

$$\|P\|_2 = \|E_l \Sigma_{l-1} + \Sigma_{l-1} E_l^\top\|_2 \quad (17)$$

$$\leq \|E_l \Sigma_{l-1}\|_2 + \|\Sigma_{l-1} E_l^\top\|_2 \quad (18)$$

$$\leq 2\|E_l\|_2 \|\Sigma_{l-1}\|_2, \quad (19)$$

where we used the triangle inequality and sub-multiplicativity of the spectral norm. Therefore, up to the second-order remainder already stated in Proposition 1, the spectral deviation from the uniformly scaled profile is controlled linearly by $\|E_l\|_2$. In the mid-depth near-identity regime where $|\rho_l|$ is small and $\|E_l\|_2$ is modest under the pooled surrogate, this bound is consistent with limited per-layer spectral drift (i.e., an approximately stable spectrum up to near-uniform scaling). $\square$

B.4. Heuristic Interpretation of Rotational Dynamics

Finally, we provide a heuristic interpretation of the small pooled update–input inner product observed in mid-depth layers. This is not a formal proof; it is a geometric intuition that links the pooled metric c_{up} to the qualitative role of the non-parallel term E_l .

Directional vs. Radial Effects Under Pooling. A small pooled update–input inner product suggests that, under the pooled metric, the dominant first-order change is not purely radial (i.e., not primarily aligned with the current pooled representation). One may therefore heuristically view the residual effect as inducing primarily directional adjustments rather than uniform scaling, but we do not claim a strict skew-symmetry property for E_l .

Implication for the Main Text Interpretation. Under the first-order decomposition in Eq. 11, this intuition is consistent with the view that ρ_l primarily controls near-uniform scaling of the second moment, while the non-parallel component aggregated in E_l contributes to gradual orientation changes of the covariance geometry.

Table 8. Hyperparameter settings for LoRA and Ours on the Commonsense Reasoning.

Hyperparameter	LoRA	Ours
λ_{orth}	–	0.0001 - 0.05 - 0.0001
λ_{ca}	–	0.1
λ_{cs}	–	0.0001 - 0.05 - 0.01
Rank r		16
α		32
Dropout		0.05
Batch size		32
Micro batch size		16
Epochs		3
Learning rate		3e-4
Target module		q, k, v, up, down

C. Dataset and Hyperparameter Details

C.1. Commonsense Reasoning

We first trained the model using 170k training samples, and then evaluated the fine-tuned model on seven commonsense reasoning question-answering benchmarks. These benchmark datasets are designed to assess models’ inference skills through complex and nuanced question sets. Hyperparameters are provided in Table 8.

Table 9. Hyperparameter settings for LoRA and Ours on arithmetic reasoning.

Hyperparameter	LoRA	Ours
λ_{orth}	–	0.0001 - 0.05 - 0.0001
λ_{ca}	–	0.1
λ_{cs}	–	0.0001 - 0.05 - 0.01
Rank r		16
α		32
Dropout		0.05
Batch size		32
Micro batch size		16
Epochs		3
Learning rate		3e-4
Cutoff length		256
Validation set size		120
Target module		q, k, v, up, down

- **OBQA** (Mihaylov et al., 2018b): a dataset composed of questions that demand multi-step reasoning, external commonsense knowledge, and deep reading comprehension. The test set comprises 500 items.
- **ARC-c/e** (Clark et al., 2018a): includes both Challenge and Easy subsets from the ARC benchmark, featuring genuine grade-school-level multiple-choice science questions. The test sets contain 2376 and 1172 questions, respectively.
- **WinoG. (WinoGrande)** (Sakaguchi et al., 2020): a binary-choice task that assesses commonsense reasoning by requiring models to choose the correct completion for a given sentence. It includes 1267 test samples.
- **PIQA** (Bisk et al., 2019): consists of physically grounded commonsense questions, where each item offers two plausible solutions. There are 1830 samples in the test set.
- **BoolQ** (Clark et al., 2019a): a binary question answering benchmark consisting of naturally occurring yes/no questions derived from unprompted queries. The test set contains 3,270 samples.
- **HellaS. (HellaSwag)** (Zellers et al., 2019a): focuses on commonsense natural language inference, providing a context and several possible sentence completions. The test set includes 10,042 instances.

C.2. Arithmetic Reasoning

For arithmetic reasoning evaluation, we consider five benchmarks. These datasets span a wide range of mathematical problems, from basic arithmetic operations to complex multi-step reasoning tasks. The model was fine-tuned on 10K training samples to comprehensively assess its numerical reasoning capabilities. Hyperparameter settings are provided in Table 9.

- **AddSub** (Hosseini et al., 2014a): a dataset composed of addition and subtraction-based word problems for elementary-level learners. It includes 395 test samples.
- **SVAMP** (Patel et al., 2021b): single-variable arithmetic word problems adapted from an existing dataset by applying minor textual modifications, aimed at up-to-4th grade students. It includes 1,000 problems in the test set.
- **AQuA** (Ling et al., 2017a): a benchmark of algebraic word problems accompanied by natural language rationales, designed to evaluate both answer accuracy and reasoning explainability. The test set includes 254 questions.
- **MultiArith** (Roy & Roth, 2015a): math word problems that require multiple arithmetic operations and reasoning steps to solve. There are 600 problems in the test set.
- **GSM8K** (Cobbe et al., 2021a): a collection of high-quality, linguistically diverse math word problems designed by human authors, targeting grade school level arithmetic reasoning. The test set contains 1,319 questions.

Table 10. Empirical results on Commonsense Reasoning. Each score denotes the mean and standard deviation (in parentheses) across five random seeds.

Method	OBQA	ARC-e	ARC-c	WinoG.	PIQA	BoolQ	HellaS.	Avg.
Llama3 + LoRA	84.9 (± 0.6)	90.2 (± 0.2)	79.9 (± 0.6)	85.5 (± 0.3)	87.9 (± 0.6)	69.9 (± 0.9)	94.4 (± 0.2)	84.7 (± 0.2)
+ Ours	85.3 (± 0.4)	90.3 (± 0.1)	80.5 (± 0.4)	85.1 (± 0.7)	88.7 (± 0.5)	71.8 (± 0.8)	94.9 (± 0.1)	85.2 (± 0.2)
Qwen2 + LoRA	89.3 (± 0.6)	92.7 (± 0.7)	83.1 (± 0.4)	85.5 (± 0.4)	89.5 (± 0.3)	73.9 (± 0.6)	94.6 (± 0.1)	86.9 (± 0.3)
+ Ours	89.1 (± 0.4)	93.2 (± 0.2)	84.2 (± 0.4)	85.7 (± 0.6)	89.8 (± 0.4)	74.1 (± 0.3)	94.8 (± 0.2)	87.3 (± 0.2)
Mistral + LoRA	84.0 (± 0.6)	85.7 (± 0.6)	73.4 (± 0.6)	83.0 (± 0.6)	87.4 (± 0.7)	71.0 (± 1.0)	90.3 (± 0.3)	82.1 (± 0.5)
+ Ours	85.4 (± 0.7)	86.1 (± 0.4)	74.6 (± 0.6)	84.5 (± 0.4)	87.5 (± 0.6)	71.8 (± 0.4)	90.7 (± 0.9)	82.9 (± 0.6)
Phi-3.5 + LoRA	88.6 (± 0.7)	94.4 (± 0.3)	85.4 (± 0.3)	83.0 (± 0.6)	86.1 (± 0.6)	70.3 (± 0.4)	90.8 (± 0.3)	85.5 (± 0.3)
+ Ours	89.3 (± 0.5)	94.7 (± 0.3)	86.0 (± 0.3)	83.6 (± 0.3)	86.8 (± 0.3)	70.5 (± 0.4)	90.8 (± 0.4)	86.0 (± 0.4)

Table 11. Empirical results on Arithmetic reasoning. Each score denotes the mean and standard deviation (in parentheses) across five random seeds.

Method	SVAMP	AddSub	AQuA	MultiArith	GSM8K	Avg.
Llama3 + LoRA	82.7 (± 0.5)	92.9 (± 1.5)	32.3 (± 0.9)	98.0 (± 0.3)	72.9 (± 0.7)	75.8 (± 0.6)
+ Ours	83.8 (± 0.3)	92.9 (± 1.0)	34.3 (± 0.5)	98.3 (± 0.3)	73.6 (± 0.5)	76.6 (± 0.4)
Qwen2 + LoRA	82.8 (± 0.7)	91.6 (± 0.4)	36.9 (± 1.1)	97.3 (± 0.7)	73.9 (± 0.5)	76.5 (± 0.7)
+ Ours	83.5 (± 0.4)	91.2 (± 0.4)	37.6 (± 0.4)	97.3 (± 0.3)	75.0 (± 0.6)	76.9 (± 0.6)
Mistral + LoRA	65.3 (± 1.5)	87.9 (± 0.7)	23.7 (± 2.0)	95.0 (± 0.6)	62.6 (± 0.6)	66.9 (± 1.5)
+ Ours	66.8 (± 1.2)	88.9 (± 0.7)	24.7 (± 1.6)	95.3 (± 0.4)	63.2 (± 0.4)	67.8 (± 1.0)
Phi-3.5 + LoRA	76.4 (± 0.3)	86.1 (± 0.7)	33.6 (± 0.7)	98.1 (± 0.4)	76.0 (± 0.2)	74.0 (± 0.3)
+ Ours	76.9 (± 0.3)	85.5 (± 0.8)	34.1 (± 0.4)	98.4 (± 0.4)	77.0 (± 0.4)	74.4 (± 0.3)

D. Unembedding-Subspace Access Metric Details

This appendix specifies the *unembedding-subspace access* metric used to operationalize *accessibility* in the main text. The key idea is to measure how strongly a layer representation couples to a fixed *readout-relevant* subspace induced by the output head, since final token logits are computed through this unembedding map.

D.1. Readout-Relevant Subspace from the Unembedding Head

Let $W_U \in \mathbb{R}^{|V| \times d}$ denote the unembedding (LM head) matrix of the *base* backbone.² To obtain a tractable estimate of the dominant readout-relevant directions, we sample m token rows from W_U to form $W_s \in \mathbb{R}^{m \times d}$ and compute the top- k right-singular subspace:

$$W_s^\top W_s = Q \Lambda Q^\top, \quad B_{\text{out}} = Q_{[:,1:k]} \in \mathbb{R}^{d \times k}, \quad k = 64, \quad (20)$$

with the corresponding orthogonal projector

$$P_{\text{out}} = B_{\text{out}} B_{\text{out}}^\top. \quad (21)$$

We use $k = 64$ throughout for simplicity; in preliminary checks the qualitative layer-wise trends were stable for $k \in \{32, 64, 128\}$.

Why the Subspace is Fixed Across Models. In our setting, the LM head is kept frozen (no updates to W_U) while training adapters/interventions. Regardless, we intentionally compute B_{out} once from the base model and hold it fixed when comparing the baseline vs. internalized backbones, so that the metric reflects *representation-readout coupling changes* rather than changes in head geometry.

²For models with tied embeddings, W_U denotes the tied matrix used for both input embedding and unembedding.

D.2. Row Sampling Protocol (Reproducible)

We sample m token indices uniformly at random from the vocabulary, excluding special tokens (BOS/EOS/PAD and other control tokens) and fixing the sampling seed. Unless stated otherwise, we use the same m and the same seed for all models and evaluations, so B_{out} and P_{out} are identical across comparisons.

D.3. Layer Representation and Pooling Protocol

For an input sequence with hidden states $H^{(l)} \in \mathbb{R}^{T \times d}$ at layer l , we form a pooled vector

$$h^{(l)} := \phi(H^{(l)}) \in \mathbb{R}^d, \quad (22)$$

using the same pooling operator ϕ as in the main text (e.g., last-token pooling). This keeps the access metric consistent with the pooled protocol used for near-identity indicators and second-moment analyses.

D.4. Access Ratio Definition and Estimation

The layer-wise unembedding-subspace access ratio is defined as the fraction of representation energy lying in the fixed readout-relevant subspace:

$$A_l = \frac{\mathbb{E} [\|P_{\text{out}} h^{(l)}\|_2^2]}{\mathbb{E} [\|h^{(l)}\|_2^2]}, \quad k = 64. \quad (23)$$

Given a prompt set $\{x_n\}_{n=1}^N$, we estimate

$$\hat{A}_l = \frac{\frac{1}{N} \sum_{n=1}^N \|P_{\text{out}} h_n^{(l)}\|_2^2}{\frac{1}{N} \sum_{n=1}^N \|h_n^{(l)}\|_2^2}, \quad (24)$$

and report $\hat{A}_l$ per layer for both the baseline and internalized (“+Ours”) models under the same prompts/tokenization/max length/pooling. We summarize the change as

$$\Delta A_l = \hat{A}_l^{\text{ours}} - \hat{A}_l^{\text{base}}. \quad (25)$$

Interpretation and Limitation. A_l measures *geometric coupling* to dominant directions of the output head (a readout-relevant subspace). It is not a token-frequency-weighted gradient attribution metric; it intentionally abstracts away from per-token logit gradients and captures only subspace-level alignment/energy.

Computation. Applying the projector is efficient via $P_{\text{out}} h = B_{\text{out}}(B_{\text{out}}^\top h)$, so evaluation overhead beyond forward passes is negligible.

E. Additional Empirical Results

E.1. Full Benchmark Results with Standard Deviations

Tables 10 and 11 present the full per-dataset empirical results on commonsense reasoning and arithmetic reasoning benchmarks. Each score indicates the mean and standard deviation computed over five random seeds.

F. Detailed Implementation of the Diagnostic Module

In this section, we provide the technical specifications of the Depth-Aware Diagnostic Intervention (DDI) module, whose high-level architecture is illustrated in Figure 3. We employ DDI as a temporary diagnostic route intended to (1) expose mid-depth representations to the terminal readout during training and (2) test whether the backbone can internalize this access pathway without retaining the auxiliary module at inference time.

F.1. Hierarchical Aggregation Architecture

To effectively capture layer-wise representational transitions, DDI employs a two-stage hierarchical aggregation mechanism.

Input Formulation. Given input hidden states $H_t = [H_t^1, \dots, H_t^L] \in \mathbb{R}^{L \times D}$ from L layers, we preserve the multi-head structure of the backbone. The input is reshaped to $\mathbb{R}^{L \times K \times d_K}$ to allow each attention head to learn distinct layer-wise access patterns, maintaining the modularity of the original Transformer.

Local and Global Aggregation. The aggregation process is divided into two stages to capture both short-range continuity and long-range semantic abstraction:

1. **Local Aggregation:** We group layers into non-overlapping blocks (e.g., a block size of 4 for Llama-3-8B), though the specific granularity can be adapted per architecture. Within each block, a linear projection compresses adjacent layer states into a local summary $Z_t^{(g)}$, preserving local representational dynamics.
2. **Global Aggregation:** These local summaries are processed by a 1D convolution-based global aggregator. This stage integrates information across the entire depth, producing the final diagnostic vector R_t that encapsulates the latent information currently under-utilized by the final layer.

F.2. Internalization via Self-Distillation

A key design principle of our analysis is “Zero Inference Overhead”. We posit that if the mid-depth bottleneck is purely structural (access-limited) rather than informational (capacity-limited), the backbone should be able to learn to access this information without external parameters. To verify this, we employ an internalization strategy: during training, the DDI module (R_t) guides the final layer output (H_t^L), but simultaneously, the backbone is forced to mimic this guided behavior via self-distillation. Consequently, at inference time, DDI is removed, and we evaluate only the updated backbone parameters.

F.3. Diagnostic Training Objectives

The loss functions are designed not to maximize metric scores directly, but to enforce specific structural behaviors: complementarity, alignment, and assimilation.

Orthogonal Loss ($\mathcal{L}_{orth}$): Forcing Complementarity. To ensure that DDI acts as a probe for *missing* information rather than redundant features, we penalize the cosine similarity between the diagnostic vector R_t and the original final-layer state H_t^L :

$$\mathcal{L}_{orth} = \frac{1}{T} \sum_{t=1}^T \cos(R_t, H_t^L)^2. \quad (26)$$

This objective forces R_t to encode representational components that are **orthogonal** (complementary) to what the final layer already possesses. Crucially, since optimizing for orthogonality alone might drive R_t toward arbitrary nuisance directions, this objective is strictly co-optimized with $\mathcal{L}_{ca}$ to ensure semantic coherence.

Correlation Alignment ($\mathcal{L}_{ca}$): Ensuring Semantic Validity. While probing for new information, R_t must remain semantically valid within the model’s feature space. We maximize the cosine similarity between the mean-centered representations of R_t and H_t^L :

$$\mathcal{L}_{ca} = 1 - \frac{1}{T} \sum_{t=1}^T \cos(R_t - \mu_R, H_t^L - \mu_H). \quad (27)$$

By focusing on centered correlations, this term ensures that the probe maintains a consistent semantic direction with the backbone, preventing it from learning spurious features.

Cosine Similarity ($\mathcal{L}_{cs}$): Testing Reversibility. This is the core objective for verifying the “reversible access restriction” hypothesis. We train the backbone’s final representation H_t^L to approximate the intervention-augmented state $\tilde{H}_t$:

$$\mathcal{L}_{cs} = 1 - \frac{1}{T} \sum_{t=1}^T \cos(\text{sg}[\tilde{H}_t], H_t^L), \quad (28)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator. This objective tests whether the backbone parameters can be updated to absorb (internalize) the access pathway identified by DDI. Successful minimization of this loss implies that the structural bottleneck can be resolved within the backbone’s own parameter space.

Total Objective. The final diagnostic objective is a weighted sum:

$$\mathcal{L} = \mathcal{L}_{base} + \lambda_{orth}\mathcal{L}_{orth} + \lambda_{ca}\mathcal{L}_{ca} + \lambda_{cs}\mathcal{L}_{cs}. \quad (29)$$

Here, $\mathcal{L}_{base}$ corresponds to the original task loss (e.g., standard LM cross-entropy), ensuring that diagnostic modifications do not degrade fundamental task capabilities.

G. Validation of Diagnostic Design Configurations

We evaluate the sensitivity of DDI to key design choices: (1) the granularity of layer aggregation and (2) the contribution of each auxiliary objective used during training/internalization. Unless stated otherwise, we report performance on the finalized backbone with DDI removed at inference.

G.1. Sensitivity to Aggregation Granularity

We vary the resolution of local/global grouping used to aggregate intermediate layers. Table 12 shows that 4 local groups and 8 global groups provide the best trade-off in our setting, while heavier compression degrades performance, suggesting that overly coarse aggregation can obscure layer-specific signals.

Table 12. Ablation study on aggregation granularity. A balanced configuration (4 Local, 8 Global) yields the most effective diagnostic resolution.

#Local Groups	#Global Groups	Overlap	Avg (%)
2	16	—	76.5
4	8	—	76.6
8	4	—	76.2
16	2	—	75.7
4	14	1	76.1
8	6	4	76.3
16	3	8	75.2

G.2. Functional Roles of Auxiliary Objectives

We ablate auxiliary objectives by (i) adding each term on top of the baseline and (ii) removing each term from the full objective. Disabling $\mathcal{L}_{cs}$ corresponds to removing the internalization/self-distillation objective. Table 13 shows:

- **Internalization loss ($\mathcal{L}_{cs}$) is the main contributor:** adding $\mathcal{L}_{cs}$ yields the largest single improvement, and removing it causes the largest drop, indicating that internalization is important for transferring training-time changes into the backbone.
- **Auxiliary regularizers ($\mathcal{L}_{orth}, \mathcal{L}_{ca}$) are context-dependent:** $\mathcal{L}_{orth}$ alone provides little benefit (and can slightly reduce performance), while, in combination with the full objective, it improves the best score, suggesting a stabilizing/regularizing role rather than a standalone driver.

G.3. Qualitative Results

We conduct qualitative analyses across various models and benchmarks to gain deeper insight into the effects of DDI on representation behavior. In the following examples, correct predictions are highlighted in green, while incorrect ones are marked in red. Each case is selected to illustrate specific strengths and limitations of our method in handling semantic abstraction, multi-step reasoning, and syntactic cues.

Table 14 illustrates a limitation of semantic abstraction. While DDI tends to rely on high-level semantic cues (e.g., “Sarah is better”), the task instead requires reasoning based on functional roles within a comparative structure. The incorrect

Table 13. Comprehensive ablation study on diagnostic objectives. The comparison between single-term addition and full-term removal highlights the synergistic nature of the losses.

Configuration	Avg (%)	Δ from Base
Baseline (Llama3 + LoRA)	75.7	-
<i>(Single Addition)</i>		
w/ $\mathcal{L}_{orth}$ only	75.6	-0.1
w/ $\mathcal{L}_{ca}$ only	75.7	0.0
w/ $\mathcal{L}_{cs}$ only	76.0	+0.3
<i>(Full Configuration)</i>		
Full Objective ($\mathcal{L}_{total}$)	76.6	+0.9
<i>(Single Removal from Full)</i>		
w/o $\mathcal{L}_{orth}$	76.0	+0.3
w/o $\mathcal{L}_{ca}$	76.4	+0.7
w/o $\mathcal{L}_{cs}$	75.8	+0.1

Table 14. Qualitative results in WinoGrande benchmark.

Base model : Llama3-8B-Instruct + LoRA	
Question	Sarah was a much better surgeon than Maria so _ always got the easier cases.
Options	Option1 : Sarah, Option2 : Maria
Original	the correct answer is option2.
Ours	the correct answer is option1.

prediction indicates that DDI’s abstraction mechanism, although effective for capturing semantic integration, may overlook shallow syntactic cues, such as role alignment and comparative reversal. These cues are essential for resolving referential ambiguity in structurally constrained contexts.

Table 15 highlights DDI’s strength in multi-step reasoning tasks that require structured abstraction. In tasks involving proportional energy distribution, temporal causality, or hypothetical scenarios, DDI consistently outperforms the original model by preserving latent reasoning chains across steps. While the original model selects a semantically plausible but incorrect answer, DDI accurately identifies the correct answer by maintaining intermediate reasoning related to kinetic energy transformation. These results support DDI’s design objective: to preserve structured semantic trajectories through selective compression, particularly in tasks that require multi-stage causal or proportional alignment.

Table 16 illustrates a representative failure case that highlights a trade-off in DDI’s abstraction strategy. In this example, the question asks for the total number of turnips, but DDI mistakenly includes pumpkins in the final answer. This suggests

Table 15. Qualitative results in ARC-C benchmark.

Base model : Phi-3.5-Mini-Instruct + LoRA	
Question	An astronaut drops a 1.0 kg object and a 5.0 kg object on the Moon. Both objects fall a total distance of 2.0 m vertically. Which of the following best describes the objects after they have fallen a distance of 1.0 m?
Answer	Answer1: They have each lost kinetic energy. Answer2: They have each gained the same amount of potential energy. Answer3: They have each lost the same amount of potential energy. Answer4: They have each gained one-half of their maximum kinetic energy.
Original	the correct answer is answer3.
Ours	the correct answer is answer4.

Table 16. Qualitative results in AddSub benchmark.

Base model : Qwen2-7B-Instruct + LoRA	
Question	Sally grew 113 turnips and 118 pumpkins . Mary grew 129 turnips . How many turnips did they grow in total?
Original	To find the total number of turnips Sally and Mary grew, we need to add the number of turnips each of them grew. Step 1: Add the number of turnips Sally grew to the number of turnips Mary grew. $113 + 129 = 242$. Step 2: The total number of turnips they grew is 242. Therefore, Sally and Mary grew a total of 242 turnips. Prediction : 242.0
Ours	Step 1: Add the number of turnips Sally grew to the number of turnips Mary grew. $113 + 129 = 242$. Step 2: Add the number of pumpkins Sally grew to the number of pumpkins Mary grew. $118 + 0 = 118$. Step 3: Add the total number of turnips to the total number of pumpkins. $242 + 118 = 360$. Therefore, they grew a total of 360 turnips. The answer in Arabic numerals is 360. Prediction : 360.0

Table 17. Qualitative results in SVAMP benchmark.

Base model : Mistral-7B-Instruct-v0.3 + LoRA	
Question	Nell collects baseball cards. She had 566 cards while Jeff had 234 cards. She gave some of her cards to Jeff and now has 535 cards left. How many cards does Jeff have now?
Original	Step 1: Find the total number of cards Nell and Jeff had before Nell gave some to Jeff. $566 + 234 = 800$. Step 2: Find the number of cards Nell gave to Jeff. $800 - 535 = 265$. Step 3: Add the number of cards Nell gave to Jeff to the number of cards Jeff had before. $234 + 265 = 499$. Therefore, Jeff now has 499 cards. The answer in Arabic numerals is 499. Prediction : 499.0
Ours	Nell gave away $566 - 535 = 31$ cards to Jeff. So, Jeff now has $234 + 31 = 265$ cards. Therefore, Jeff has 265 cards now. The answer in Arabic numerals is 265. Prediction : 265.0

that although DDI supports conceptual integration, it can blur task-specific word boundaries when the question requires strict attention to surface-level details. The example emphasizes a key limitation of abstraction-based models: while helpful for reasoning tasks that need high-level understanding, abstraction must be carefully controlled when accurate lexical interpretation is important.

Table 17 presents a case where DDI demonstrates its strength in handling multi-step reasoning. The task requires understanding a hidden quantity transfer over time and correctly updating another person’s state based on it. While the original model incorrectly sums the total number of cards and fails to isolate the relevant change, DDI correctly infers the number of cards transferred and applies that change to Jeff’s count. This example highlights how DDI helps maintain structured reasoning over sequential steps and improves performance on tasks that involve causal or temporal dependencies.

G.4. Depth-Wise Representation Improvements Across Models

Figures 15 to 20 present the effective rank and log-determinant of covariance across depth for Qwen2, Phi-3.5, and Mistral models. In all cases, DDI consistently demonstrates improved depth-wise representation structure compared to the base models. Specifically, DDI tends to suppress redundancy in the middle layers while enhancing representational richness near the output. This pattern aligns with our findings on Llama3 and further confirms the generalizability of DDI’s design across different model architectures.

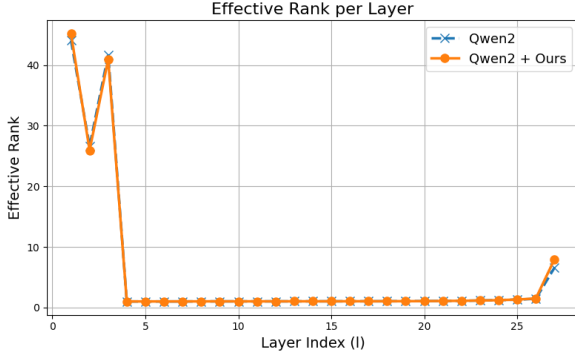


Figure 15. Layer-wise Effective Rank comparison between Qwen2 and Qwen2+Ours.

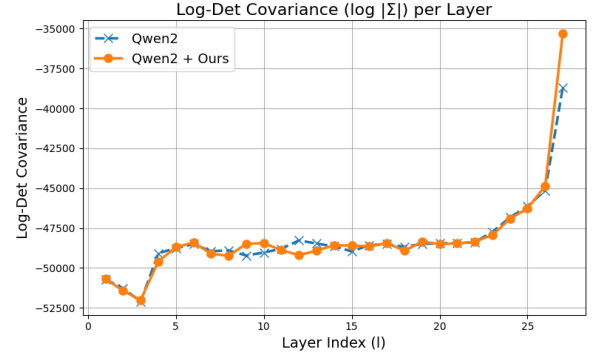


Figure 16. Log-Det second-moment matrix across Layers in Qwen2 vs. Qwen2+Ours.

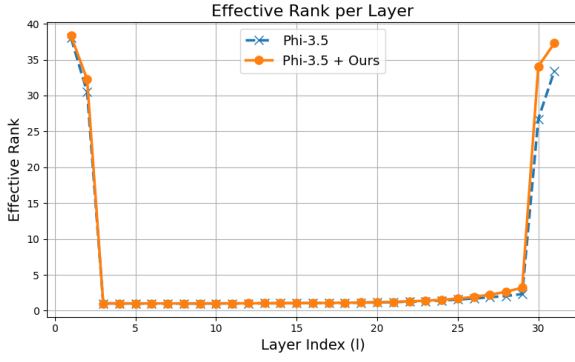


Figure 17. Layer-wise Effective Rank comparison between Phi-3.5 and Phi-3.5+Ours.

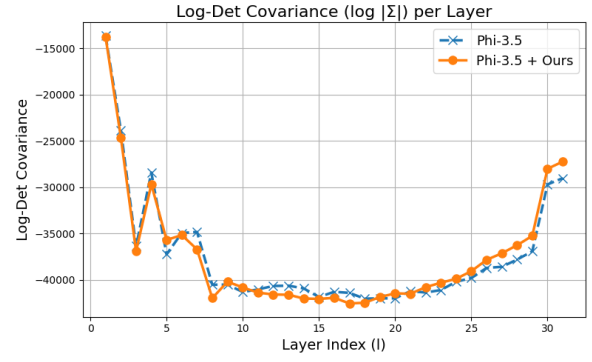


Figure 18. Log-Det second-moment matrix across Layers in Phi-3.5 vs. Phi-3.5+Ours.

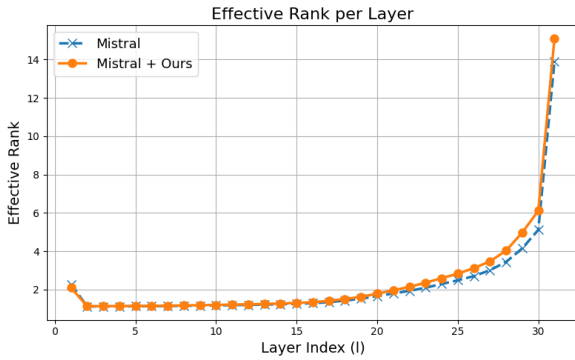


Figure 19. Layer-wise Effective Rank comparison between Mistral and Mistral+Ours.

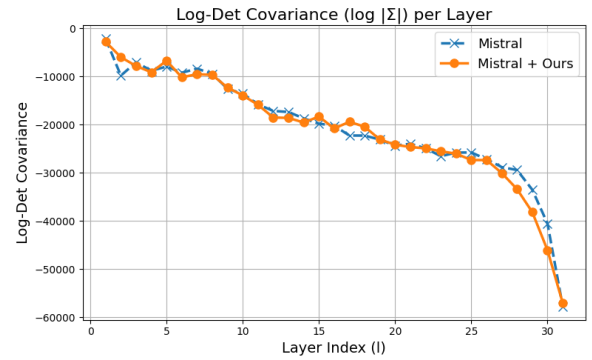


Figure 20. Log-Det second-moment matrix across Layers in Mistral vs. Mistral+Ours.